Skin and subcutaneous tissue disorders:
Angioedema, eczema, erythema, pruritus, hyperhidrosis, urticaria, drug eruption, toxic skin eruption, rash

Muscoloskeletal, connective tissue and bone disorders:
Arthralgia, back pain, muscle spasms (cramps in legs), pain in extremity (leg pain), myalgia, tendon pain (tendinitis like symptoms)

Renal and urinary disorders:
Renal impairment including acute renal failure

General disorders and administration site conditions:
Chest pain, Influenza-like illness, asthenia (weakness)

Investigations:
Haemoglobin decreased, blood uric acid increased, blood creatinine increased, hepatic enzyme increased, blood creatine phosphokinase (CPK) increased

USE IN SPECIAL POPULATIONS

- **Pregnancy**

USE IN PREGNANCY
When used in pregnancy during second and third trimesters, drug that act directly on rennin angiotensin system can cause injury and even death to the developing fetus. When pregnancy is detected, telmisartan should be discontinued as soon as possible (see **CONTRAINDICATIONS** and **WARNINGS AND PRECAUTIONS**).

There is no adequate information from the use of telmisartan in pregnant women. In animals reproductive toxicity have been reported.

Epidemiological evidence regarding the risk of teratogenicity following exposure to ACE inhibitors during the first trimester of pregnancy has not been conclusive; however a small increase in risk cannot be excluded. Whilst there is no controlled epidemiological information on the risk with angiotensin II receptor antagonists, similar risks may exist for this class of drugs.

Unless continued angiotensin II receptor antagonist therapy is considered essential, patients planning pregnancy should be changed to alternative antihypertensive treatments which have an established safety profile for use in pregnancy. When pregnancy is diagnosed, treatment with angiotensin II receptor antagonists should be stopped immediately, and, if appropriate, alternative therapy should be started.

Exposure to angiotensin II receptor antagonist therapy during the second and third trimesters is known to induce human fetotoxicity (decreased renal function, oligohydramnios, skull ossification retardation) and neonatal toxicity (renal failure, hypotension, hyperkalaemia).

Should exposure to angiotensin II receptor antagonists have occurred from the second trimester of pregnancy, ultrasound check of renal function and skull is recommended.

Infants whose mothers have taken angiotensin II receptor antagonists should be closely observed for hypotension.

- **Lactation**

Because no information is available regarding the use of telmisartan during breast-feeding, telmisartan is not recommended and alternative treatments with better established safety profiles during breast-feeding are preferable, especially while nursing a newborn or preterm infant.

OVERDOSAGE

There is limited information available with regard to overdose in humans.

Symptoms: The most prominent manifestations of telmisartan overdose were hypotension and tachycardia; bradycardia dizziness, increase in serum creatinine, and acute renal failure have also been reported.

Treatment: Telmisartan is not removed by haemodialysis. The patient should be closely monitored, and the treatment should be symptomatic and supportive. Management depends on the time since ingestion and the severity of the symptoms. Suggested measures include induction of emesis and / or gastric lavage. Activated charcoal may be useful in the treatment of overdosage. Serum electrolytes and creatinine should be monitored frequently. If hypotension occurs, the patient should be placed in a supine position, with salt and volume replacement given quickly.

STORAGE : Store below 30°C, protect from moisture.

KEEP ALL MEDICINES OUT OF REACH OF CHILDREN

PACKAGING: Blister Strips of 10 x 10's and 3 x 10's

Date of Revision: August 2018

Manufactured By /
Product Registration Holder :
RANBAXY
(MALAYSIA) SDN. BHD.
Lot 23, Bakar Arang Industrial Estate,
08000 Sungai Petani, Kedah.
Malaysia.

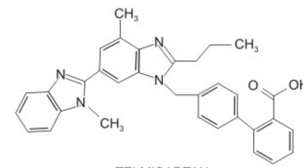
XXXXXX

For the use only of a Registered Medical Practitioner

PRESCRIBING INFORMATION
TELEACT
(Telmisartan Tablets 40/80mg)

COMPOSITION
Each tablet contains:
Telmisartan ....40/80 mg

DESCRIPTION
TELEACT contains telmisartan, which is a non-peptide angiotensin II receptor (type AT_1) antagonist. Telmisartan is chemically described as 4'-[(1, 4'-dimethyl-2'-propyl [2,6'-bi-1H-benzimidazol]-1'-yl)methyl]-[1,1'-biphenyl]-2-carboxylic acid. Its empirical formula is $C_{33}H_{30}N_4O_2$, its molecular weight is 514.63 and its structural formula is:



TELMISARTAN

PRODUCT DESCRIPTION

For 40 mg strength: Off-white to light yellow coloured, oblong tablets, debossed with 'T12' on one side and plain on the other side
For 80 mg strength: Off-white to light yellow coloured, oblong tablets, debossed with 'T13' on one side and plain on the other side

PHARMACODYNAMIC AND PHARMACOKINETIC PROPERTIES

- **Pharmacodynamics**

Telmisartan is an orally active and specific angiotensin II receptor (type AT_1) antagonist. Telmisartan displaces angiotensin II with very high affinity from its binding site at the AT_1 receptor subtype, which is responsible for the known actions of angiotensin II. Telmisartan does not exhibit any partial agonist activity at the AT_1 receptor. Telmisartan selectively binds the AT1 receptor. The binding is long-lasting. Telmisartan does not show affinity for other receptors, including AT_2 and other less characterised AT receptors. The functional role of these receptors is not known, nor is the effect of their possible over stimulation by angiotensin II, whose levels are increased by telmisartan. Plasma aldosterone levels are decreased by telmisartan. Telmisartan does not inhibit human plasma renin or block ion channels. Telmisartan does not inhibit angiotensin converting enzyme (kininase II), the enzyme which also degrades bradykinin. Therefore it is not expected to potentiate bradykinin-mediated adverse effects.

In human, an 80 mg dose of telmisartan almost completely inhibits the angiotensin II evoked blood pressure increase. The inhibitory effect is maintained over 24 hours and still measurable up to 48 hours.

- **Pharmacokinetics**

Absorption of telmisartan is rapid although the amount absorbed varies. The mean absolute bioavailability for telmisartan is about 50 %. When telmisartan is taken with food, the reduction in the area under the plasma concentration-time curve ($AUC_{0-\infty}$) of telmisartan varies from approximately 6 % (40 mg dose) to approximately 19 % (160 mg dose). By 3 hours after administration, plasma concentrations are similar whether telmisartan is taken fasting or with food.

The small reduction in AUC is not expected to cause a reduction in the therapeutic efficacy. There is no linear relationship between doses and plasma levels. C_{max} and to a lesser extent AUC increase disproportionately at doses above 40 mg.

Telmisartan is largely bound to plasma protein (>99.5 %), mainly albumin and alpha-1 acid glycoprotein. The mean steady state apparent volume of distribution (V_{dss}) is approximately 500 L.

Telmisartan is metabolised by conjugation to the glucuronide of the parent compound. No pharmacological activity has been shown for the conjugate.

Telmisartan is characterised by biexponential decay pharmacokinetics with a terminal elimination half-life of >20 hours. The maximum plasma concentration (C_{max}) and, to a smaller extent, the area under the plasma concentration-time curve (AUC), increase disproportionately with dose. There is no evidence of clinically relevant accumulation of telmisartan taken at the recommended dose. Plasma concentrations were higher in females than in males, without relevant influence on efficacy.
After oral (and intravenous) administration, telmisartan is nearly exclusively excreted with the faeces, mainly as unchanged compound. Cumulative urinary excretion is <1 % of dose. Total plasma clearance (Cl_{tot}) is high (approximately 1,000 ml/min) compared with hepatic blood flow (about 1,500 ml/min).

Special Populations

Gender effects:
Differences in plasma concentrations has been reported, with C_{max} and AUC being approximately 3- and 2-fold higher, respectively, in females compared to males.

Elderly patients:
The pharmacokinetics of telmisartan does not differ between the elderly and those younger than 65 years.

Patients with renal impairment:
In patients with mild to moderate and severe renal impairment, doubling of plasma concentrations has been reported. However, lower plasma concentrations were reported in patients with renal insufficiency undergoing dialysis. Telmisartan is highly bound to plasma protein in renal-insufficient patients and cannot be removed by dialysis. The elimination half-life is not changed in patients with renal impairment.

Patients with hepatic impairment:
In patients with hepatic impairment an increase in absolute bioavailability up to nearly 100 % has been reported. The elimination half-life is not changed in patients with hepatic impairment.

INDICATIONS / USAGE

Hypertension
Treatment of essential hypertension in adults

DOSE AND METHOD OF ADMINISTRATION
TELEACT is available as 40/80mg Tablets. It is not suitable for patients requiring 20 mg telmisartan. Therefore, other suitable available strengths of telmisartan should be used in such cases.
Treatment of essential hypertension:
The usually effective dose is 40 mg once daily. Some patients may already benefit at a daily dose of 20 mg. In cases where the target blood pressure is not achieved, the dose of telmisartan can be increased to a maximum of 80 mg once daily. Alternatively, telmisartan may be used in combination with thiazide-type diuretics such as hydrochlorothiazide, which has been shown to have an additive blood pressure lowering effect with telmisartan. When considering raising the dose, it must be borne in mind that the maximum antihypertensive

effect is generally attained four to eight weeks after the start of treatment. In patients with severe hypertension treatment with telmisartan at doses up to 160 mg alone and in combination with hydrochlorothiazide 12.5 - 25 mg daily was well tolerated and effective.

Telmisartan may be taken with or without food.

Special patient populations:

Renal impairment: No dosage adjustment is required for patients with mild to moderate renal impairment, including those on haemodialysis.

Limited information is available in patients with severe renal impairment or haemodialysis. A lower starting dose of 20 mg is recommended in these patients (see **WARNINGS AND PRECAUTIONS**).

Hepatic impairment: In patients with mild to moderate hepatic impairment, the dose should not exceed 40 mg once daily (see **WARNINGS AND PRECAUTIONS**).

Elderly: No dose adjustment is necessary for elderly patients.

Paediatric patients: Telmisartan is not recommended for use in children below 18 years due to a lack of data on safety and efficacy.

CONTRAINDICATIONS

- Hypersensitivity to the active substance or to any of the excipients
- Second and third trimesters of pregnancy (see sections **WARNINGS AND PRECAUTIONS** and **USE IN SPECIAL POPULATIONS**)
- Lactation
- Biliary obstructive disorders
- Severe hepatic impairment

WARNINGS AND PRECAUTIONS

Pregnancy:
Angiotensin II receptor antagonists should not be initiated during pregnancy. Unless continued angiotensin II receptor antagonist therapy is considered essential, patients planning pregnancy should be changed to alternative antihypertensive treatments which have an established safety profile for use in pregnancy. When pregnancy is diagnosed, treatment with angiotensin II receptor antagonists should be stopped immediately, and, if appropriate, alternative therapy should be started (see **CONTRAINDICATIONS and USE IN SPECIAL POPULATIONS**).

Hepatic impairment:
Telmisartan is not to be given to patients with cholestasis, biliary obstructive disorders or severe hepatic impairment (see **CONTRAINDICATIONS**) since telmisartan is mostly eliminated with the bile. These patients can be expected to have reduced hepatic clearance for telmisartan. Telmisartan should be used only with caution in patients with mild to moderate hepatic impairment.

Renovascular hypertension:
There is an increased risk of severe hypotension and renal insufficiency when patients with bilateral renal artery stenosis or stenosis of the artery to a single functioning kidney are treated with medicinal products that affect the renin-angiotensin-aldosterone system.

Renal impairment and kidney transplantation:
When telmisartan is used in patients with impaired renal function, periodic monitoring of potassium and creatinine serum levels is recommended. There is no experience regarding the administration of telmisartan in patients with recent kidney transplantation.

Intravascular hypovolaemia:
Symptomatic hypotension, especially after the first dose of telmisartan, may occur in patients who are volume and/or sodium depleted by vigorous diuretic therapy, dietary salt restriction, diarrhoea, or vomiting. Such conditions should be corrected before the administration of telmisartan. Volume and/or sodium depletion should be corrected prior to administration of telmisartan.

Dual blockade of the renin-angiotensin-aldosterone system:
As a consequence of inhibiting the renin-angiotensin-aldosterone system, hypotension, syncope, hyperkalaemia, and changes in renal function (including acute renal failure) have been reported in susceptible individuals, especially if combining medicinal products that affect this system. Dual blockade of the renin-angiotensin-aldosterone system (e.g. by adding an ACE-inhibitor to an angiotensin II receptor antagonist) is therefore not recommended in patients with already controlled blood pressure and should be limited to individually defined cases with close monitoring of renal function.

Other conditions with stimulation of the renin-angiotensin-aldosterone system:
In patients whose vascular tone and renal function depend predominantly on the activity of the renin-angiotensin-aldosterone system (e.g. patients with severe congestive heart failure or underlying renal disease, including renal artery stenosis), treatment with medicinal products that affect this system such as telmisartan has been associated with acute hypotension, hyperazotaemia, oliguria, or rarely acute renal failure (see **SIDE EFFECTS/ ADVERSE REACTIONS**).

Primary aldosteronism:
Patients with primary aldosteronism generally will not respond to antihypertensive medicinal products acting through inhibition of the renin-angiotensin system. Therefore, the use of telmisartan is not recommended.

Aortic and mitral valve stenosis, obstructive hypertrophic cardiomyopathy:
As with other vasodilators, special caution is indicated in patients suffering from aortic or mitral stenosis, or obstructive hypertrophic cardiomyopathy.

Hyperkalaemia:
The use of medicinal products that affect the renin-angiotensin-aldosterone system may cause hyperkalaemia.

In the elderly, in patients with renal insufficiency, in diabetic patients, in patients concomitantly treated with other medicinal products that may increase potassium levels, and/or in patients with intercurrent events, hyperkalaemia may be fatal.

Before considering the concomitant use of medicinal products that affect the renin-angiotensin-aldosterone system, the benefit risk ratio should be evaluated.

The main risk factors for hyperkalaemia to be considered are:
- Diabetes mellitus, renal impairment, age (>70 years)

- Combination with one or more other medicinal products that affect the renin-angiotensin-aldosterone system and/or potassium supplements. Medicinal products or therapeutic classes of medicinal products that may provoke hyperkalaemia are salt substitutes containing potassium, potassium-sparing diuretics, ACE inhibitors, angiotensin II receptor antagonists, non steroidal anti-inflammatory medicinal products (NSAIDs, including selective COX-2 inhibitors), heparin, immunosuppressives (cyclosporin or tacrolimus), and trimethoprim.

- Intercurrent events, in particular dehydratation, acute cardiac decompensation, metabolic acidosis, worsening of renal function, sudden worsening of the renal condition (e.g. infectious diseases), cellular lysis (e.g. acute limb ischemia, rhabdomyolysis, extend trauma).

Close monitoring of serum potassium in at risk patients is recommended (see **DRUG INTERACTIONS**).

This medicinal product contains sorbitol. Patients with rare hereditary problems of fructose intolerance should not take telmisartan.

Ethnic differences:
As reported for angiotensin converting enzyme inhibitors, telmisartan and the other angiotensin II receptor antagonists are apparently less effective in lowering blood pressure in black people than in non-blacks, possibly because of higher prevalence of low-renin states in the black hypertensive population.

Other:
As with any antihypertensive agent, excessive reduction of blood pressure in patients with ischaemic cardiopathy or ischaemic cardiovascular disease could result in a myocardial infarction or stroke.

Fetal/Neonatal Morbidity and Mortality
Drugs that act directly on the renin-angiotensin system can cause fetal and neonatal morbidity and death when administered to pregnant women. Several dozen cases have been reported in the world literature in patients who were taking angiotensin converting enzyme inhibitors. When pregnancy is detected, discontinue telmisartan as soon as possible.

The use of drugs that act directly on the renin-angiotensin system during the second and third trimesters of pregnancy has been associated with fetal and neonatal injury, including hypotension, neonatal skull hypoplasia, anuria, reversible or irreversible renal failure, and death. Oligohydramnios has also been reported, presumably resulting from decreased fetal renal function; oligohydramnios in this setting has been associated with fetal limb contractures, craniofacial deformation, and hypoplastic lung development. Prematurity, intrauterine growth retardation, and patent ductus arteriosus have also been reported, although it is not clear whether these occurrences were related to exposure to the drug.

These adverse effects do not appear to have resulted from intrauterine drug exposure that has been limited to the first trimester. Inform mothers whose embryos and fetuses are exposed to an angiotensin II receptor antagonist only during the first trimester that most reports of fetal toxicity have been associated with second and third trimester exposure. Nonetheless, when patients become pregnant or are considering pregnancy, have the patient discontinue the use of telmisartan as soon as possible.

Rarely (probably less often than once in every thousand pregnancies), no alternative to an angiotensin II receptor antagonist will be found. In these rare cases, the mothers should be apprised of the potential hazards to their fetuses, and serial ultrasound examinations should be performed to assess the intra-amniotic environment.

If oligohydramnios is reported, telmisartan should be discontinued unless they are considered life-saving for the mother. Contraction stress testing (CST), a non-stress test (NST), or biophysical profiling (BPP) may be appropriate, depending upon the week of pregnancy. Patients and physicians should be aware, however, that oligohydramnios may not appear until after the fetus has sustained irreversible injury.

Infants with histories of in utero exposure to an angiotensin II receptor antagonist should be closely observed for hypotension, oliguria, and hyperkalemia. If oliguria occurs, attention should be directed toward support of blood pressure and renal perfusion. Exchange transfusion or dialysis may be required as a means of reversing hypotension and/or substituting for disordered renal function.

Effect on ability to drive and use machines:
No studies on the effects on the ability to drive and use machines have been performed. However, when driving vehicles or operating machinery it should be taken into account that dizziness or drowsiness may occasionally occur when taking antihypertensive therapy.

DRUG INTERACTIONS
Telmisartan may increase the hypotensive effect of other antihypertensive agents. Other interactions of clinical significance have not been identified.

Co-administration of telmisartan did not result in clinically significant interaction with digoxin, warferin, hydrochlorthiazide, glibenclamide, ibuprofen, paracetamol, simvastatin and amlodipine.

For digoxin a 20% increase in median plasma digoxin trough concentration has been observed (39% in a single case), monitoring of plasma digoxin levels should be considered.

In a reported study co-administration of telmisartan and ramipril led to an increase of up to 2.5 fold in the AUC_{0-24} and Cmax of ramipril and ramiprilat. The clinical relevance of this observation is not known.

Reversible increases in serum lithium concentrations and toxicity have been reported during concomitant administration of lithium with angiotensin converting enzyme inhibitors. Cases have also been reported with angiotensin II receptor antagonists, including telmisartan. Therefore, serum lithium level monitoring is advisable during concomitant use.

Treatment with NSAIDs (i.e. acetylsalicylic acid (ASA) at anti-inflammatory dosage regimens, COX-2 inhibitors and non-selective NSAIDs) is associated with the potential for acute renal insufficiency in patients who are dehydrated. Compounds acting on Rennin Angiotensin System like telmisartan may have synergistic effects. Patients receiving NSAIDS and telmisartan should be adequately hydrated and be monitored for renal function at the beginning of combined treatment.

A reduced effect of antihypertensive drugs like telmisartan by inhibition of vasodilating prostaglandins has been reported during combined treatment with NSAIDs.

SIDE EFFECTS/ ADVERSE REACTIONS
The overall incidence of adverse events reported with telmisartan was usually comparable to placebo. The incidence of adverse events was not dose related and showed no correlation with gender, age or race of the patients.

Infections and infestations:
Sepsis including fatal outcome, urinary tract infection (including cystitis), upper respiratory tract infections

Blood and the lymphatic system disorders:
Anaemia, eosinophilia, thrombocytopenia

Immune system disorders:
Anaphylactic reaction, hypersensitivity

Metabolism and nutrition disorders:
Hyperkalaemia

Psychiatric disorders:
Anxiety, insomnia, depression,

Nervous system disorders:
Syncope (faint)

Eye disorders:
Visual disturbance

Ear and labyrinth disorders:
Vertigo

Cardiac disorders:
Bradycardia, tachycardia

Vascular disorders:
Hypotension, orthostatic hypotension

Respiratory disorders:
Dyspnoea

Gastrointestinal disorders:
Abdominal pain, diarrhoea, dry mouth, dyspepsia, flatulence, stomach discomfort, vomiting

Hepatobiliary disorders:
Hepatic function abnormal/liver disorder